

OTS-03 Summary Document

Overview

This document summarises the OTS-03 lecture on option pricing models and greeks. The lecture discusses the features of options, their relation with the underlying asset, put-call parity, risk neutrality and the Black-Scholes-Merton model for option pricing. In addition to this, this lecture also covers the interpretation of option Greeks.

The following topics are covered:

- Model-independent features of options
- Put-Call Parity
- Option pricing variables and parameters
- Properties of BSM solution
- Option Greeks

Model Independent features of options

- **American > European:** If you don't choose to exercise, an American option is a European option. So the American option is European plus some extra value.
- **Longer Dated American Options > Short Dated:** You can turn the longer-dated option into the short one by exercising the longer-dated American option. So it is a short-dated option plus some more optionality.
- **Call < Underlying:** If you exercise, you get the underlying asset. So if Call price > stock price, sell the call and buy the underlying asset.
- **Put < Strike:** If you exercise, you get the strike minus the current price. So the highest value the option can ever have is the strike (if $S=0$).
- **Lower Strike C > Higher Strike C:** There will be times when a lower strike call can be exercised, and the higher strike call cannot.

If these model-independent features do not adhere in the real market, there will be an arbitrage opportunity.

Put/Call Parity

- Put-call parity defines the relationship between the puts and calls. It is important for put-call parity to hold as there will be an arbitrage opportunity otherwise.
- The relationship is given as -

$$S + P = C + X e^{-rT}$$

- This equation shows that buying a stock and a put is equivalent to buying a call and a zero-coupon bond with the future value equivalent to the strike price.

The following snapshot shows the initial value and value at expiration for multiple instruments.

Instrument	Initial Value	Value at Expiration	
		If $S < X$	If $S > X$
Long P(X)	P(X)	(X-S)	0
Short C(X)	-C(X)	0	-(S-X)
Long Stock	S	S	S
Total Portfolio	$S + P(X) - C(X)$	X	X

Note

- A call is like borrowing money to buy stock and a put, i.e. it is a limited downside leveraged position in the underlying.
- The payoff from any instrument can be replicated with the other three.
- For call and put to have equal value, the strike price must be the forward price.
- The relationship does not hold for American options as we can't be sure of the duration of a short option position.
- If we can make a bullish options position into a bearish options position by adding stock, then we should also be able to neutralize an option directionally.
- So P/C parity is an important precursor to volatility trading.

Option Pricing: Inputs

- Underlying price.
- Strike: The strike price of the options contract.
- Interest rates: Rates are the discount factor for cash flows.
- Dividends: Income from the stock which affects its future value.
- Time.
- Volatility.

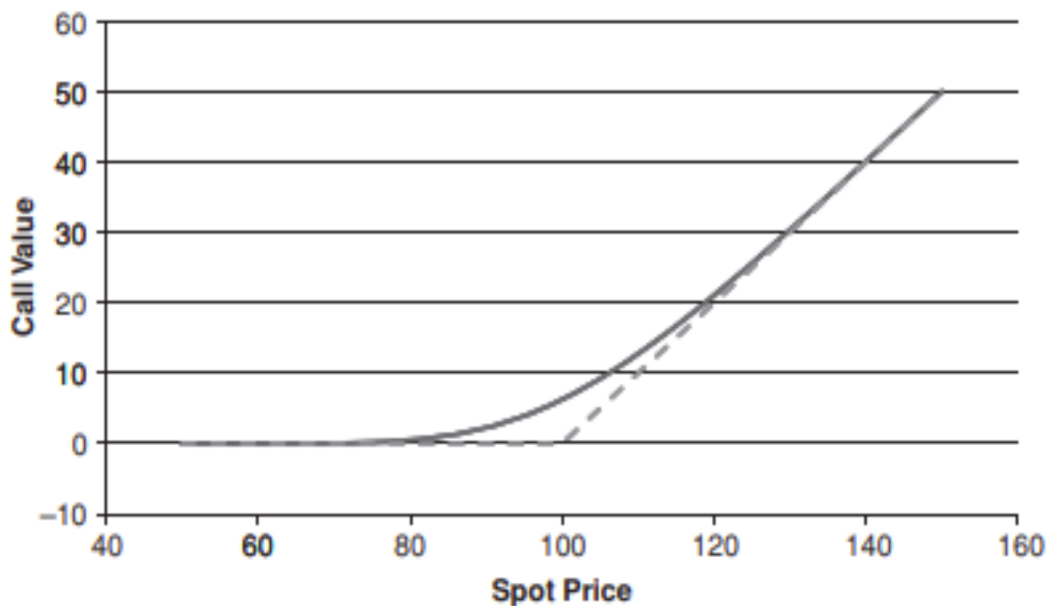
Black Scholes Model

The Black Scholes model provides the analytical framework for valuing options. It is the most commonly used option pricing model. It takes into account the strike price, the stock price, time to expiry, volatility and the interest rate and gives the option value as an output.

Solution for European Options

The option price is essentially an interpolation between the stock price and the bond price.

The following snapshot shows the BSM solutions for European options for a 100 strike call



Option value

- The option value is given by -

$$C = SN(d_1) - Xexp(-rT)N(d_2)$$

- $N()$ is the cumulative normal distribution.

$$d_1 = \frac{\ln\left(\frac{S}{X}\right) + \left(r + \frac{\sigma^2}{2}\right)T}{\sigma\sqrt{T}}$$

$$d_2 = d_1 - \sigma\sqrt{T}$$

where,

S is the underlying stock price,

X is the strike price,

r - risk-free interest rate,

T - time to expiration,

σ - Standard deviation of log returns.

- A call is a part ($N(d_1)$) of the stock and a part ($-e^{-rT}N(d_2)$) of the cash/bond.
- $N(d_1)$ is the amount of stock we are exposed to.
- $N(d_2)$ is the amount of cash we are exposed to, i.e. what we need to deliver at the expiration
- This interpretation is only true in a risk-neutral world; a world that doesn't exist.
- In the real world, the probabilities are mainly driven by drift, something we have no interest in (as a pricing variable) or significant ability to predict.

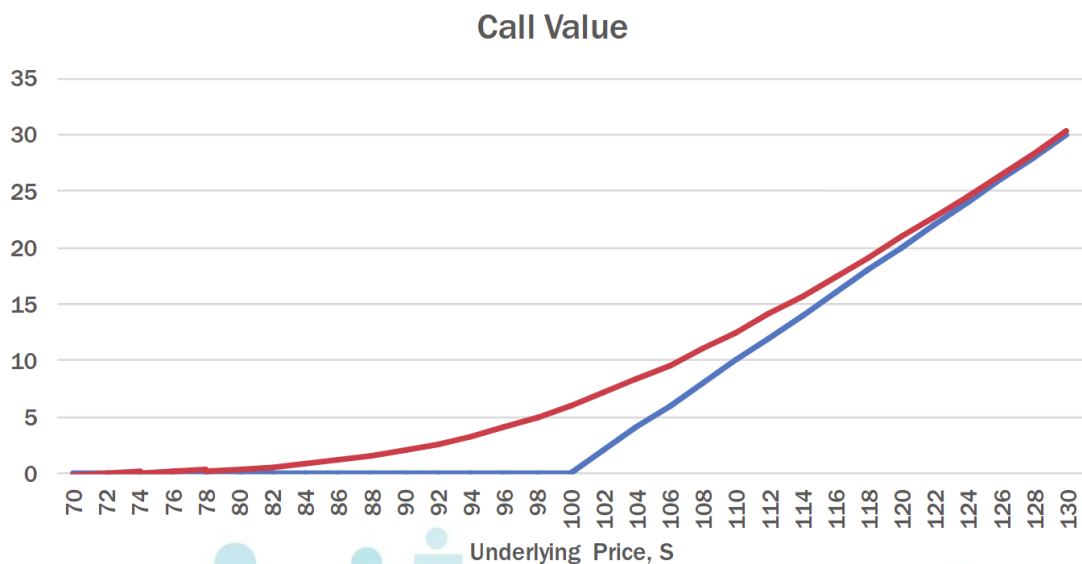
Greeks

- "Greeks" is the collective term for partial derivatives (in the mathematical sense) of the option price. Volatility traders generally quantify their positions in terms of greek exposure. Since the BSM equation is linear, greeks can be aggregated across a position by adding the greeks of each option.

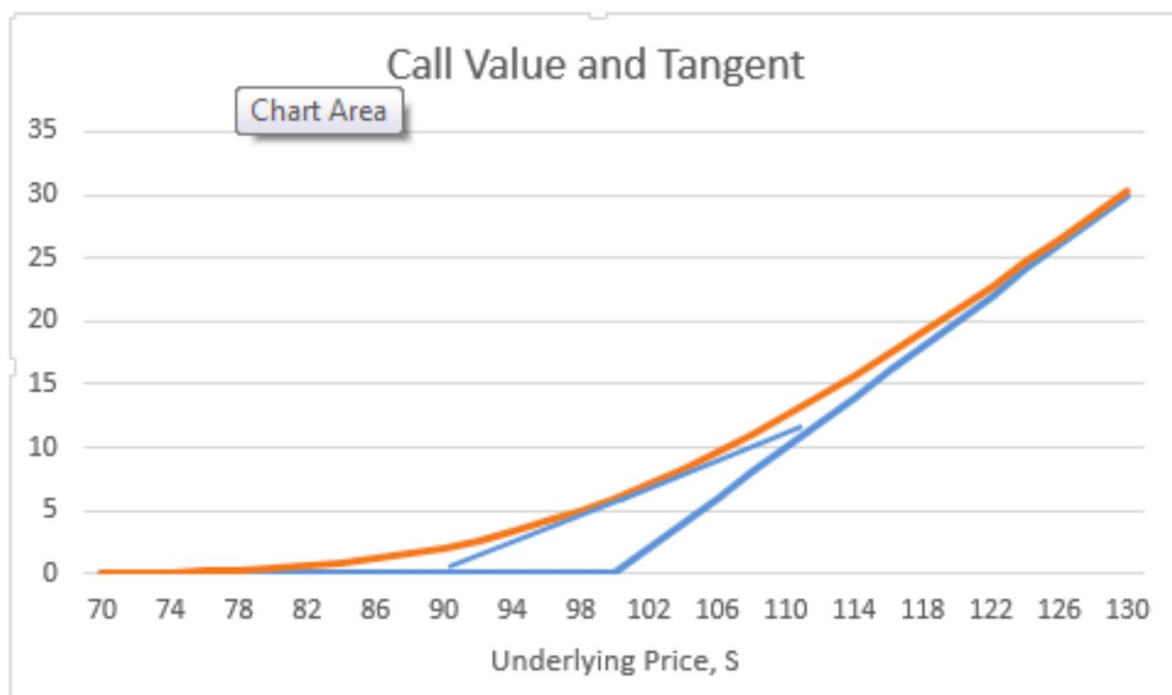
Delta

- Delta is the partial derivative of the option price w.r.t. underlying.
- Delta is the tangent of the option price vs the underlying graph

The following snapshot shows the expiration value (blue) and current premium (red) for a call with a strike price of 100

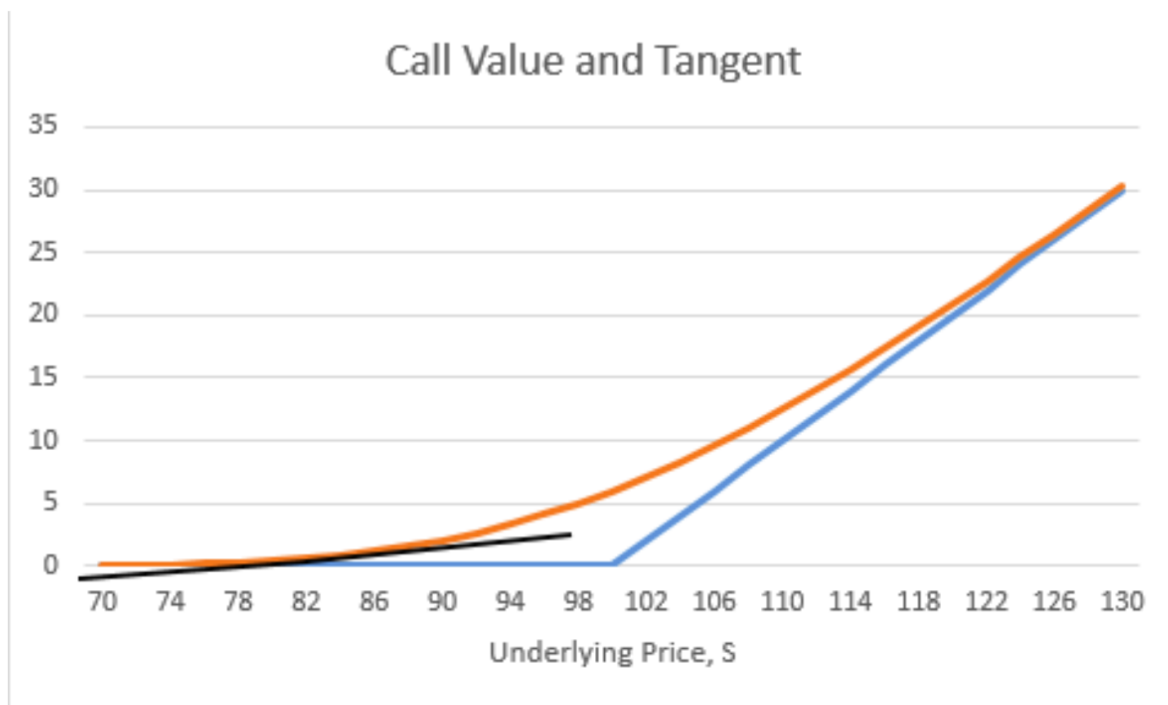


The following snapshot shows the tangent of a call value.



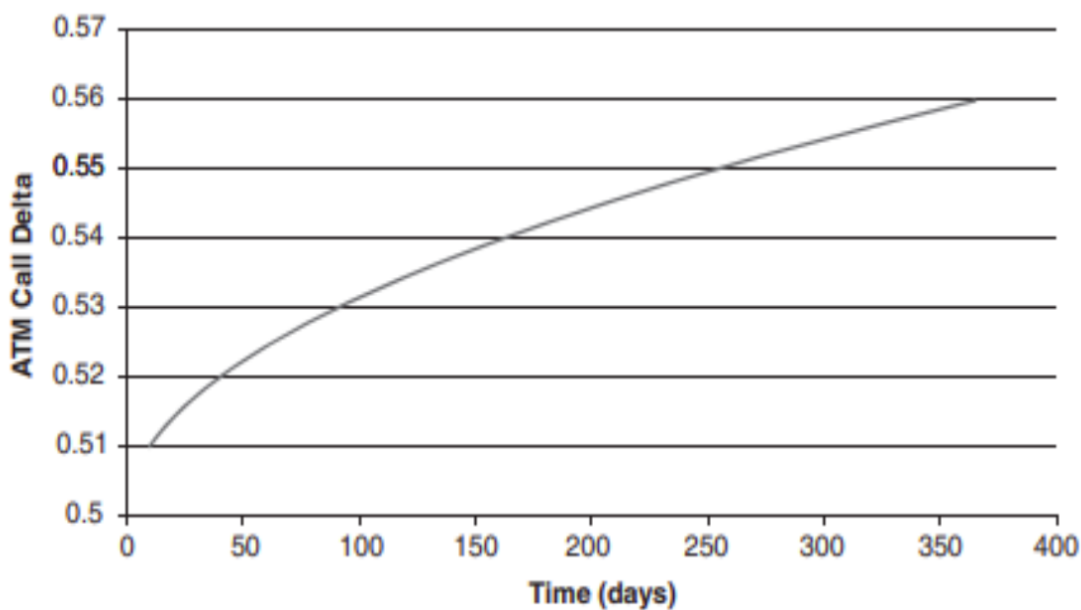
- If the underlying price is much lower than the strike, the option is way out-of-the-money, and the delta will be almost zero.

The following snapshot shows the delta line when $S=80$



Delta vs time

The following snapshot shows the ATM call delta value over time.



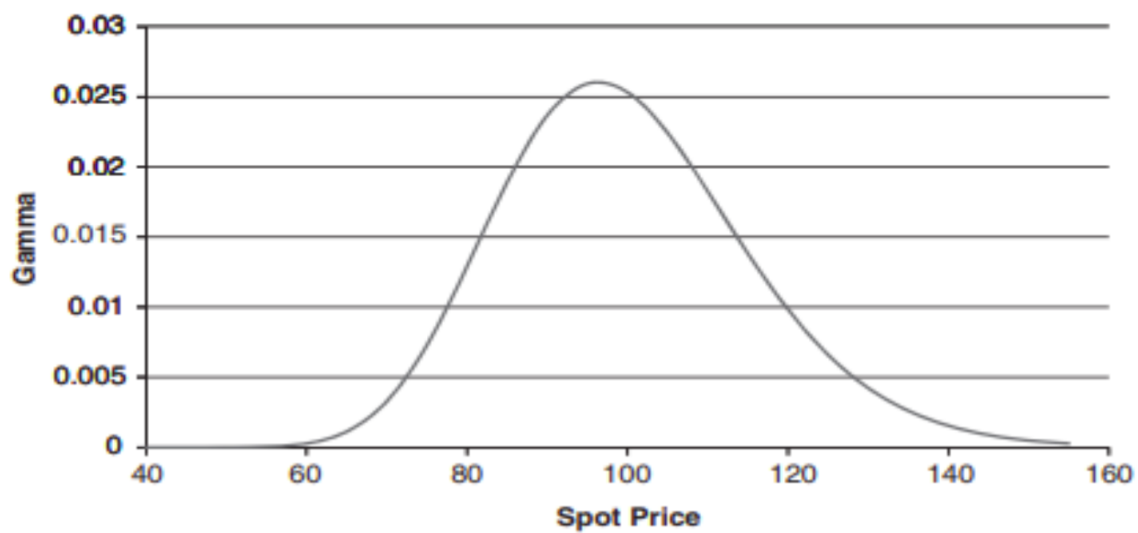
- At the infinite time (or volatility) call option delta goes to one and put option delta to zero.
- At high volatility, the stock can go much higher but is bounded below as it can't go below 0.

Gamma

- Gamma is the partial derivative of the delta w.r.t. the underlying. Simply put, gamma is the rate of change of delta. To manage delta, you must know gamma. Gamma is highest for ATM short-dated options.

Gamma vs Spot

The following snapshot shows the plot with spot price on the x-axis and the respective gamma values on the y-axis for a 100 strike call.

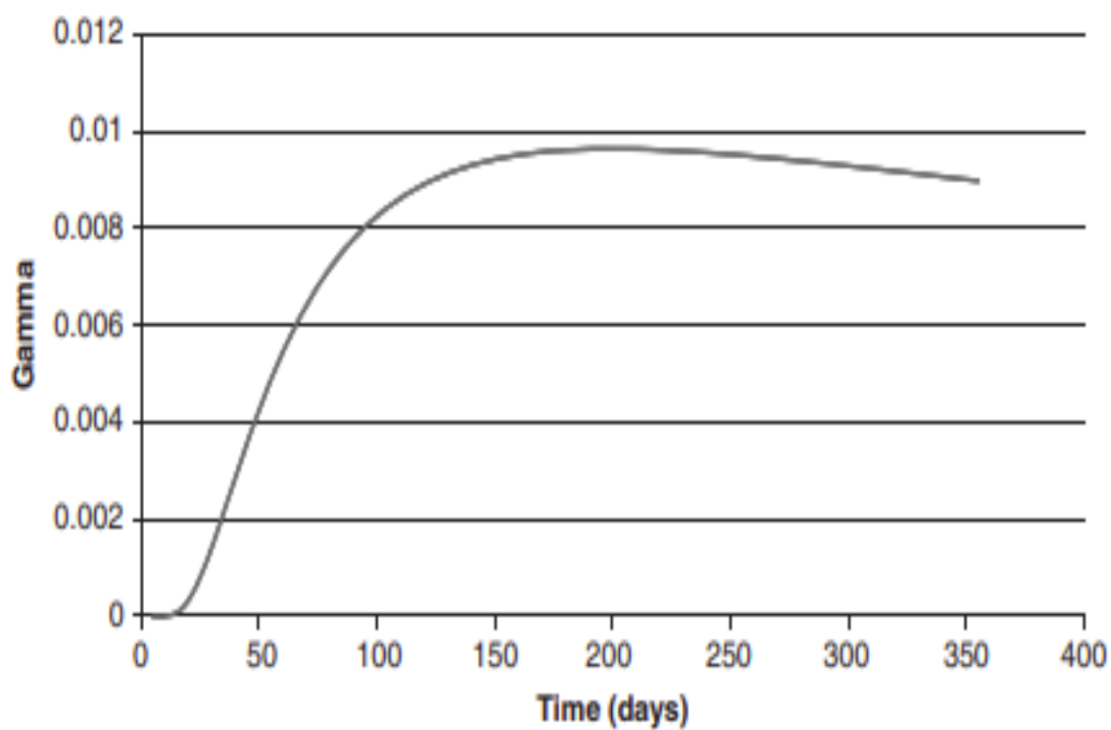


- Unsurprisingly, delta looks a lot like a cumulative normal distribution, and gamma looks a lot like a normal distribution.

Gamma vs Time

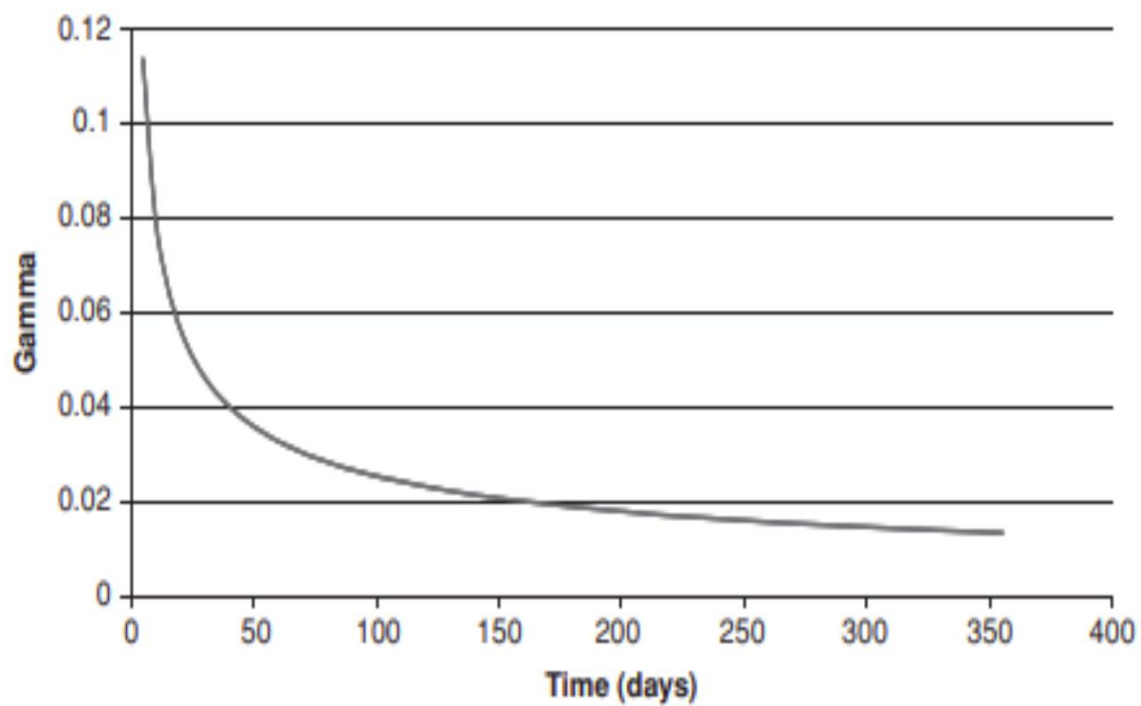
Out-of-the-money option

The following snapshot shows the gamma vs time plot of out-of-the-money options.

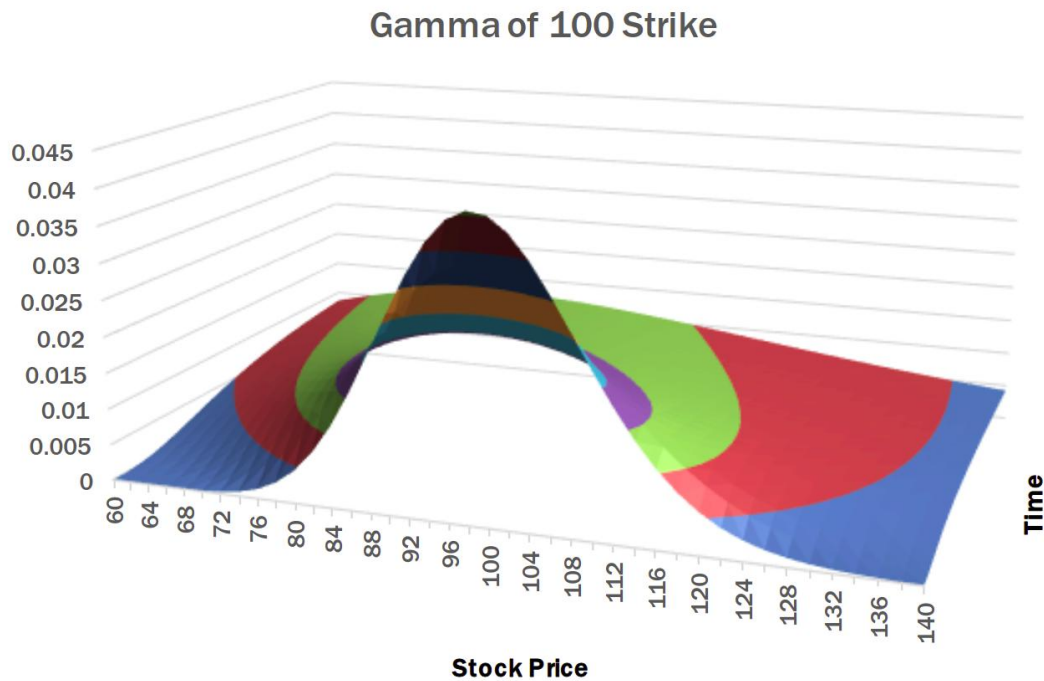


At-the-money option

The following snapshot shows the gamma vs time plot of at-the-money options.



The following snapshot shows the 3D plot with the price on the x-axis, gamma on the y-axis and time on the z-axis.



Theta

- Theta is the partial derivative of the option price w.r.t. time.
- The formula is given by -

$$\theta_{call} = -\frac{S\sigma n(d_1)}{2\sqrt{t}} - rXexp(-rT)N(d_2)$$

$$\theta_{put} = -\frac{S\sigma n(d_1)}{2\sqrt{t}} + rXexp(-rT)N(-d_2)$$

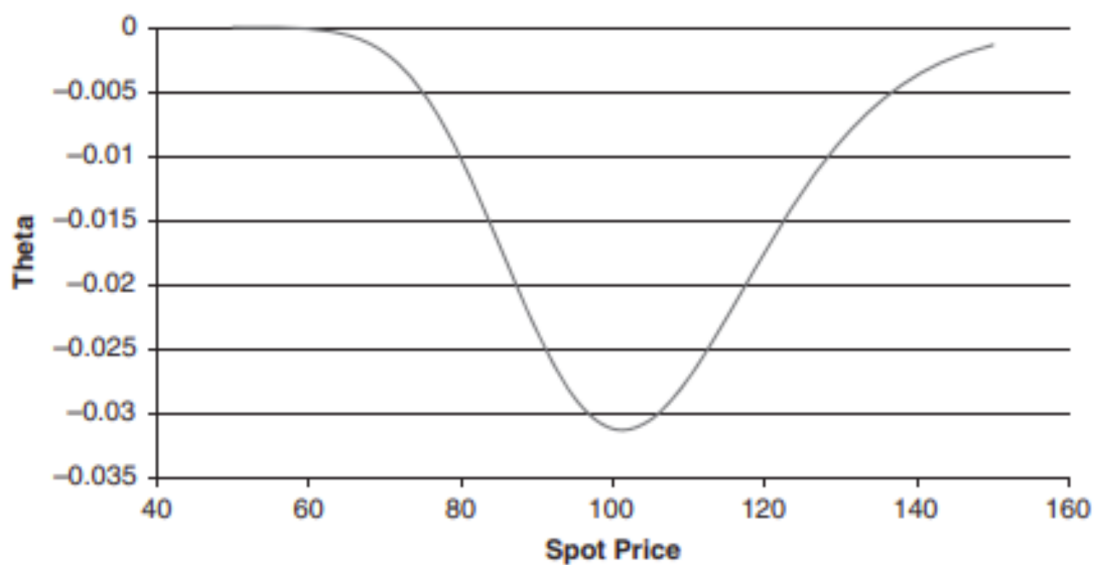
- The first term (same for puts and calls) is the time decay, the amount the option declines because volatility has less time to act.
- The second term is often misunderstood (or just ignored).

$$\theta_{call} - \theta_{put} = rXexp(-rT)$$

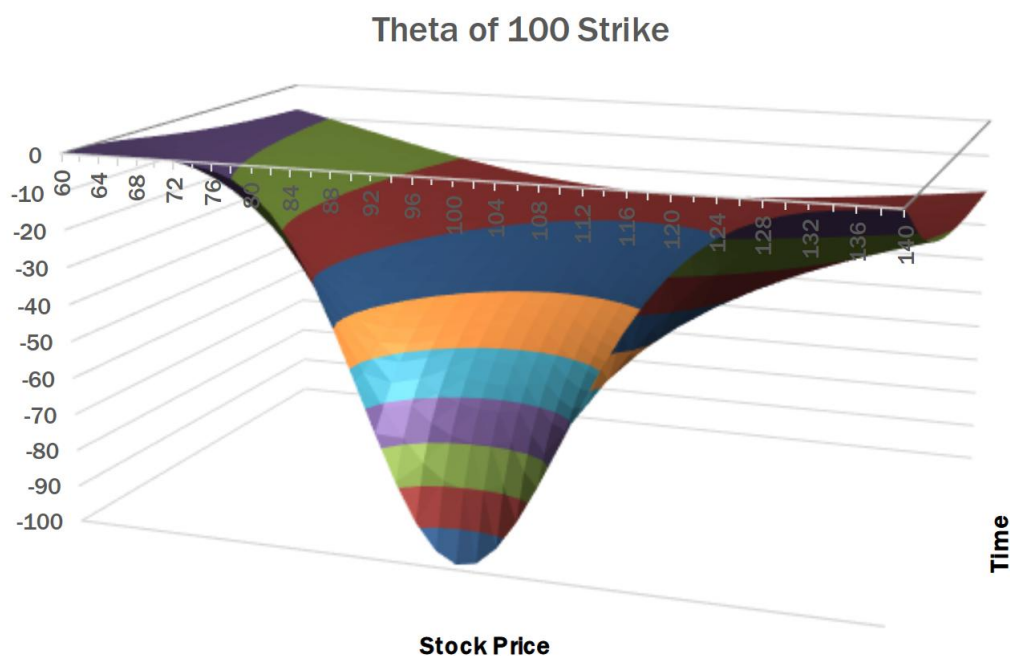
- From the Put-call parity discussion, we know a long call and short put has no volatility exposure. However, it does have to carry the cost of holding the strike value in cash.
- Theta allocates this carry across calls and puts according to the chance of each finishing in the money.

Theta vs Spot

The following snapshot shows the plot with spot price on the x-axis and the respective theta values on the y-axis for a 100 strike call.



The following snapshot shows the 3D plot with the price on the x-axis, theta on the y-axis and time on the z-axis.

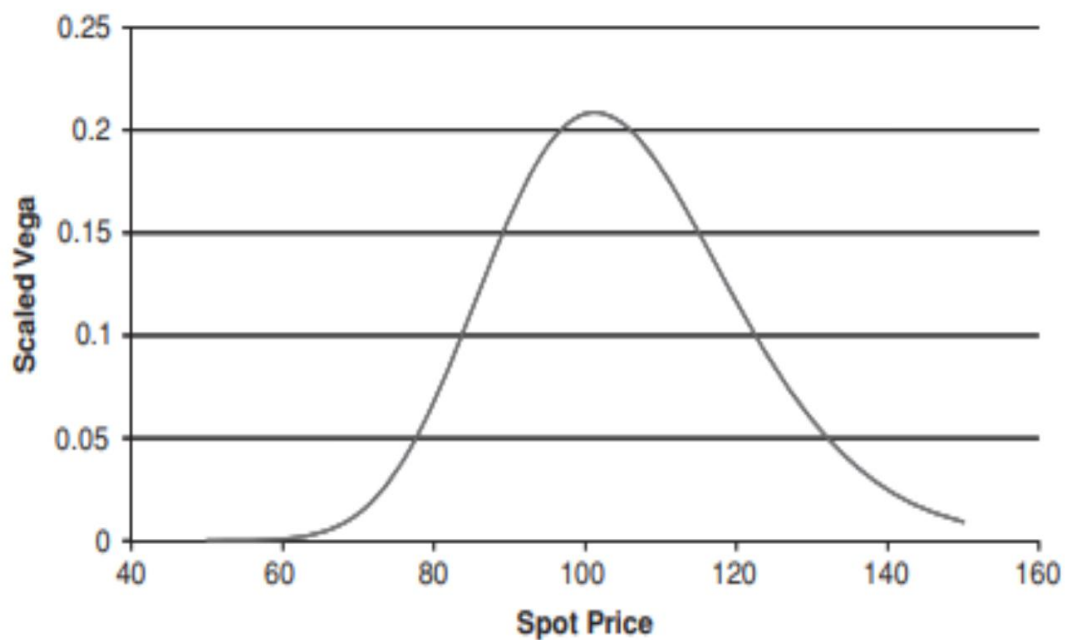


Vega

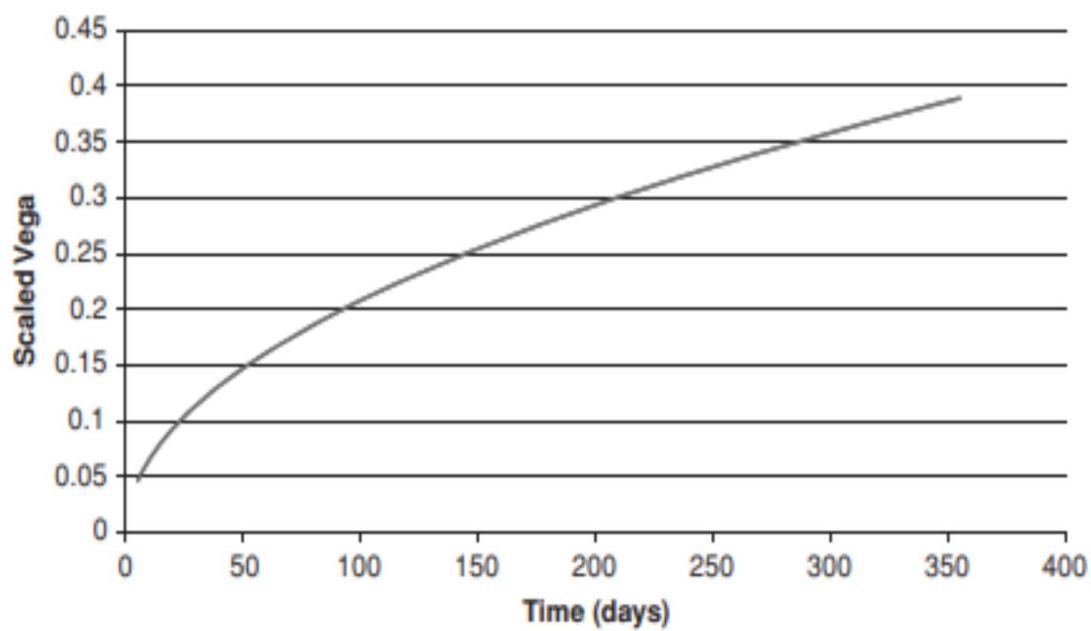
- Vega is the partial derivative of the option price w.r.t. volatility.
- It's inconsistent with the assumption of constant volatility but very important to traders.

$$Vega_{call} = Sn(d_1)\sqrt{t} = Vega_{put}$$

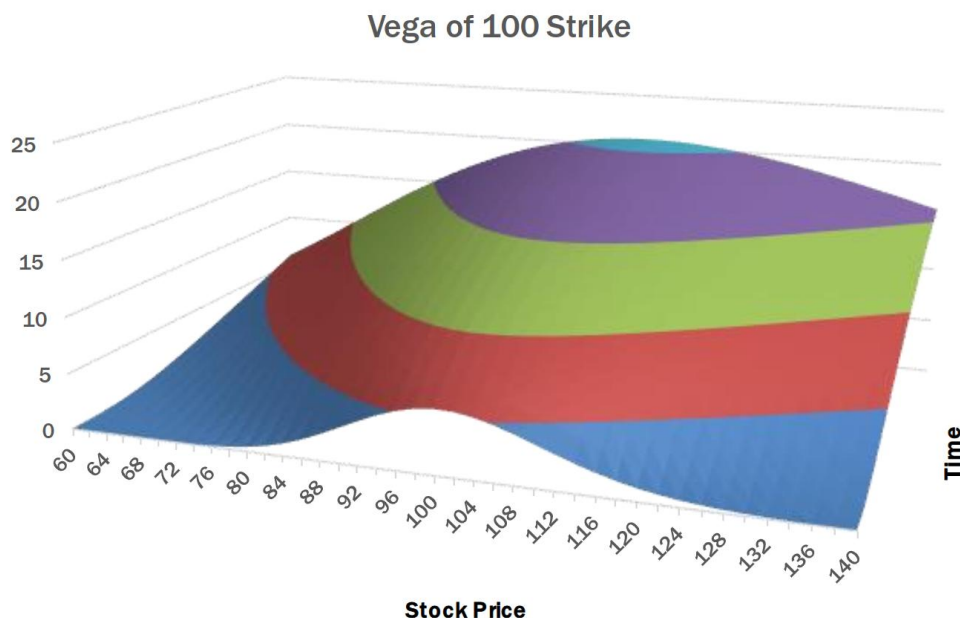
The following snapshot shows the plot with spot price on the x-axis and the respective vega values on the y-axis for a 100 strike call.



The following snapshot shows the plot with time on the x-axis and the respective vega values on the y-axis.



The following snapshot shows the 3D plot with the price on the x-axis, vega on the y-axis and time on the z-axis.



Minor greeks: Rho

- Rho is the partial derivative of the option price w.r.t. interest rates.
- It's inconsistent with the assumption of constant rates.
- Usually ignored by traders, as it is managed at the firm level by the treasury.
- Generally scaled to be dollar change for a 1% move in rates.

$$\rho_c = TX \exp(-rT) N(d_2)$$

$$\rho_p = -TX \exp(-rT) N(-d_2)$$

Secondary greeks

- From just looking at equations or graphs it is clear that the Greeks also have derivatives, e.g. the derivative of delta w.r.t. volatility, the derivative of Vega with respect to time, etc.
- Any professional-level trading or risk system will show these greeks, but a detailed understanding of their characteristics isn't essential.
- There is no standard nomenclature for these derivatives, e.g. the derivative of the delta with respect to volatility is variously called "Vanna", "DdeltaDvol", "DdelV" or even "alpha", depending on who the trader learned from.

Trading screen

The following snapshot shows the trading screen with bid-ask quotes for options.

DEC 31 '20

JAN 08 '21

JAN 15 '21

JAN 22 '21

MORE ▾

14 DAYS

22 DAYS

29 DAYS

36 DAYS

TABBED VIEW ▾

All STRIKES ▾

SMART ▾

UVXY ▾

100

IV: 127.2%

CALLS

VEGAIMPLIED VOL....

BID x ASK

DELTA

STRIKE

PUTS

VEGAIMPLIED VOL....

BID x ASK

DELTA

0.003	93.2%	3.65 x 3.70	0.959	7	0.003	93.1%	0.05 x 0.06	-0.041
0.006	100.6%	2.82 x 2.86	0.877	8	0.006	99.9%	0.21 x 0.22	-0.123
0.010	108%	2.17 x 2.20	0.757	9	0.010	110.1%	0.57 x 0.58	-0.243
0.011	122.2%	1.72 x 1.78	0.637	10	0.011	120.8%	1.11 x 1.12	-0.363
0.012	130.3%	1.41 x 1.42	0.536	11	0.012	131.8%	1.80 x 1.81	-0.464
0.012	140.2%	1.19 x 1.20	0.457	12	0.012	140.5%	2.56 x 2.59	-0.543
0.011	150.1%	1.03 x 1.04	0.395	13	0.011	151%	3.40 x 3.45	-0.605
0.010	159.6%	0.90 x 0.91	0.346	14	0.010	159%	4.25 x 4.30	-0.654
0.011	167.4%	0.81 x 0.82	0.308	15	0.011	166.1%	5.15 x 5.20	-0.692
0.009	175%	0.73 x 0.74	0.277	16	0.009	173.5%	6.05 x 6.15	-0.723
0.010	180%	0.66 x 0.68	0.250	17	0.010	182%	7.00 x 7.10	-0.750

ON

Strategy Builder

Strategies ▾

ACTN

RT

LST TRD DAY

STRIKE

TYPE

DELTA

THETA

BID/ASK

SIZE

Order Entry

Default ▾

LMT

STP

DAY ▾

advanced ▾

Account: ▾

Submit Order

Additional resources

- “Options, Futures and Other Derivatives” by John Hull.
- “Know Your Weapon” by Espen Haug
- “Options Trading” by Euan Sinclair